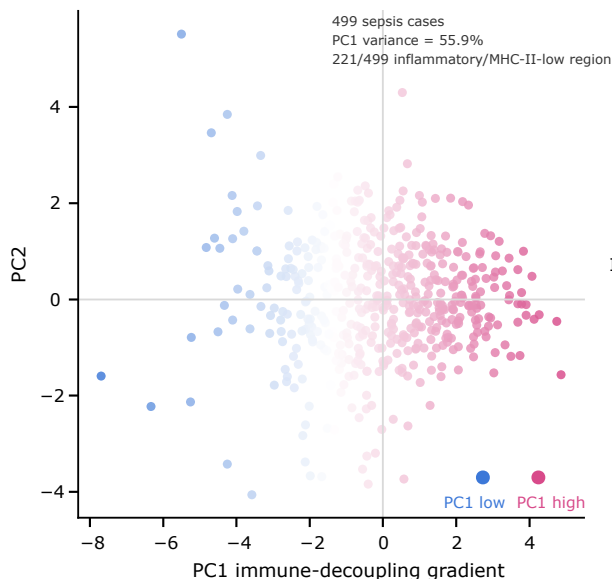
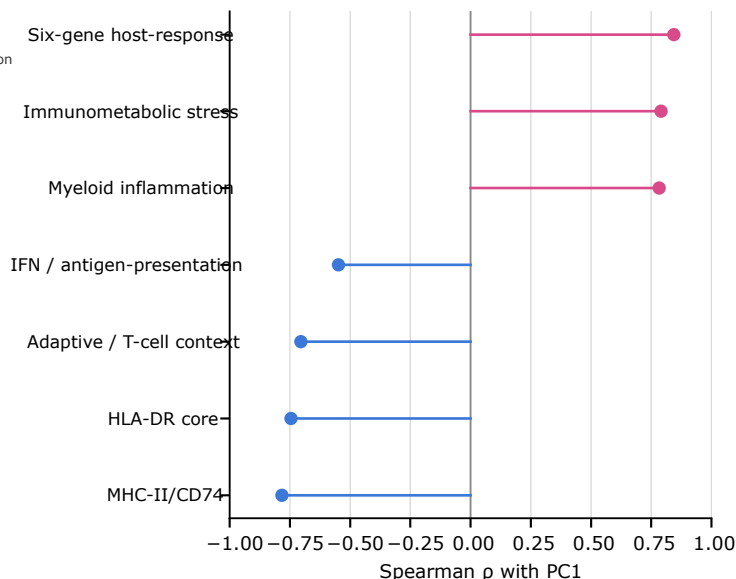
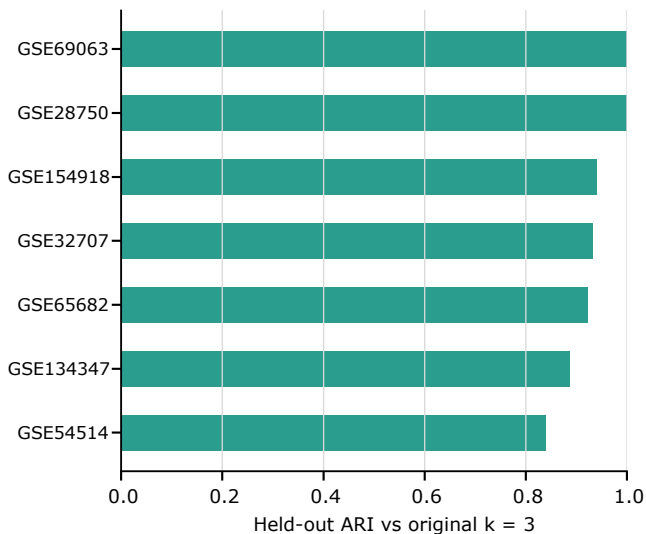


A Case-only immune-decoupling spectrum**B** Signature alignment with PC1**C** Cross-cohort transfer stability**D** Signed pathway association